

Seafood pollutant levels in Georgia

This dataset and the following information was provided by Data Classroom, Inc., and is available for free online for educational use.

Background

Glynn County, Georgia has four EPA Superfund sites—places with serious pollution that need long-term cleanup. Because these sites are close together, Brunswick (the nearby town) faces a difficult environmental challenge.

Like many coastal towns, Brunswick has a strong fishing culture. People have eaten local fish and shellfish for generations. But as pollutants entered the soil and waterways, the marine food chain began to show higher levels of mercury and PCBs (polychlorinated biphenyls).

To protect public health, the [Georgia Department of Natural Resources](#) (GDNR) tests fish around the state and publishes [seafood consumption recommendations](#). These recommendations explain how often it is safe to eat certain species from specific locations. They are based on scientific testing of pollutant levels in fish tissue, and they can change as scientists learn more or as cleanup efforts make a difference.

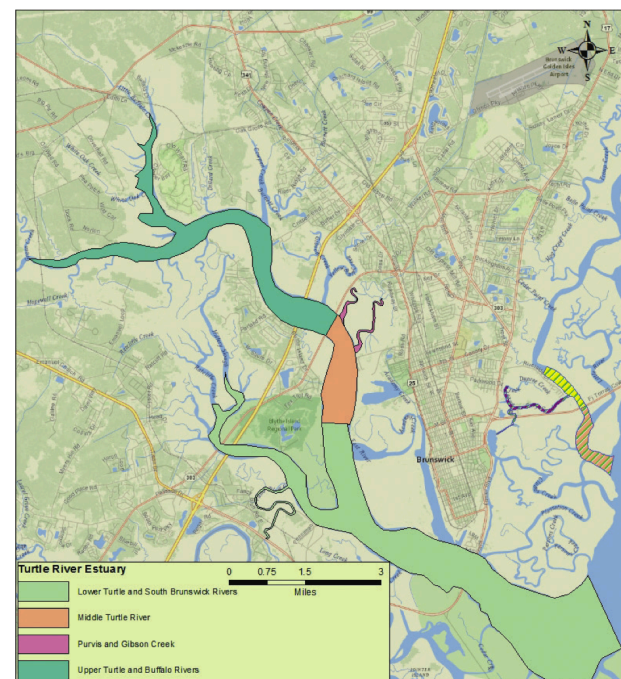
In the mid-1990s, after Glynn County sites were added to the Superfund list, the GDNR issued strict advisories for local seafood—often listed as “Do Not Eat.” The dataset you’ll explore shows how guidance for Brunswick has changed over time at three different locations as cleanup has progressed and time has passed.

Note: For real-life food decisions, always check the **most current** guidance from the GDNR. This activity uses historical data for educational purposes only.

Dataset

These data were collected from publications titled “Guidelines for Eating Fish from Georgia Waters” produced by the Georgia Department of Natural Resources Environmental Protection Division and Wildlife Resources Division. We sourced data from three years which cover the start of this data collection and recommendations (1997), the most recent data collection (2023), and a year in the middle (2010).

Recommendations are determined based on fish tissue



sampling. From the report: “Standard collections for fish consumption guidelines (FCG) are comprised of 3 composites of each species, where each composite contains 5 individual fish of that species. A sample size of fifteen fish has been found necessary to obtain a representative contaminant level to account for natural variations within a population. Composite sample analysis allows us to assess this number of fish and minimize the laboratory analytical support costs.” (You can read [more here](#))

To see the full list of publications per year, visit the [Digital Library of Georgia](#).

Variables

Location - this variable groups the location of fish sampling into waterways through and surrounding Brunswick, GA.

Year - year of publication for the consumption advisory.

Seafood Species - species for which the consumption recommendation is surveyed.

Consumption Recommendation - recommendation for how much of the seafood type should be consumed at what rate.

PCB - whether or not Polychlorinated Biphenyls (PCB's) are found in the fish tissue sample.

Mercury - whether or not mercury is present in the fish tissue sample.

Acknowledgements

The research reported in this activity was supported by Partnership for Innovation and funded by Georgia Foundation for Public Education under award Smart Communities Challenge 2023 Project.

